



Session 11B: Trigonometric Identities, Equations, and Beyond

Phase 2 — Classical Techniques | 105 min

Part A: Trigonometric Identities — Your Algebraic Arsenal

Example 1: Sum and Difference Formulas — Splitting and Merging Angles

$$\sin(A + B) = \sin A \cos B + \cos A \sin B.$$

$$\sin(A - B) = \sin A \cos B - \cos A \sin B.$$

$$\cos(A + B) = \cos A \cos B - \sin A \sin B.$$

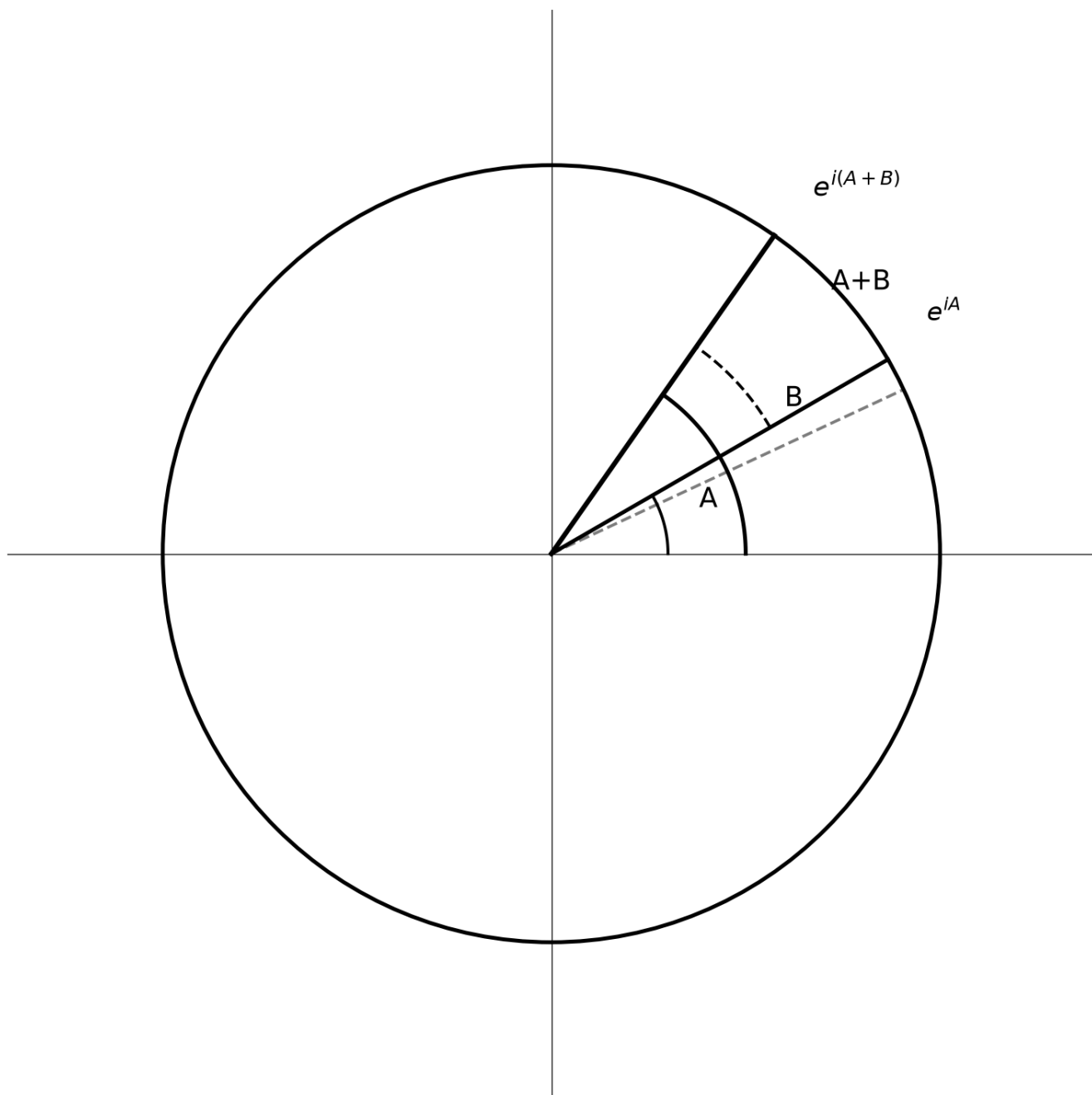
$$\cos(A - B) = \cos A \cos B + \sin A \sin B.$$

$$\tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}.$$

$$\sin 75^\circ = \sin(45^\circ + 30^\circ) = \frac{\sqrt{2}}{2} \cdot \frac{\sqrt{3}}{2} + \frac{\sqrt{2}}{2} \cdot \frac{1}{2} = \frac{\sqrt{6} + \sqrt{2}}{4}.$$

$$\cos 15^\circ = \cos(45^\circ - 30^\circ) = \frac{\sqrt{2}}{2} \cdot \frac{\sqrt{3}}{2} + \frac{\sqrt{2}}{2} \cdot \frac{1}{2} = \frac{\sqrt{6} + \sqrt{2}}{4}.$$

$$\tan 75^\circ = \frac{\tan 45^\circ + \tan 30^\circ}{1 - \tan 45^\circ \tan 30^\circ} = \frac{1 + 1/\sqrt{3}}{1 - 1/\sqrt{3}} = 2 + \sqrt{3}.$$



Geometric insight: $e^{i(A+B)} = e^{iA}e^{iB}$ — adding angles = multiplying complex numbers on the unit circle. Expand $(\cos A + i \sin A)(\cos B + i \sin B)$ and match real/imaginary parts to get all four sum formulas at once.

Example 2: Double-Angle, Half-Angle, and Triple-Angle Formulas

$$\sin 2\theta = 2 \sin \theta \cos \theta.$$

$$\cos 2\theta = \cos^2 \theta - \sin^2 \theta = 2 \cos^2 \theta - 1 = 1 - 2 \sin^2 \theta.$$

$$\tan 2\theta = \frac{2 \tan \theta}{1 - \tan^2 \theta}.$$

$$\text{Half-angle: } \sin^2 \frac{\theta}{2} = \frac{1 - \cos \theta}{2}, \cos^2 \frac{\theta}{2} = \frac{1 + \cos \theta}{2}.$$

Triple-angle:

$$\sin 3\theta = 3 \sin \theta - 4 \sin^3 \theta.$$

$$\cos 3\theta = 4 \cos^3 \theta - 3 \cos \theta.$$

$$\tan 3\theta = \frac{3 \tan \theta - \tan^3 \theta}{1 - 3 \tan^2 \theta}.$$

Example 3: Harmonic Addition — Turning $a \sin x + b \cos x$ into One Wave

$$3 \sin x + 4 \cos x.$$

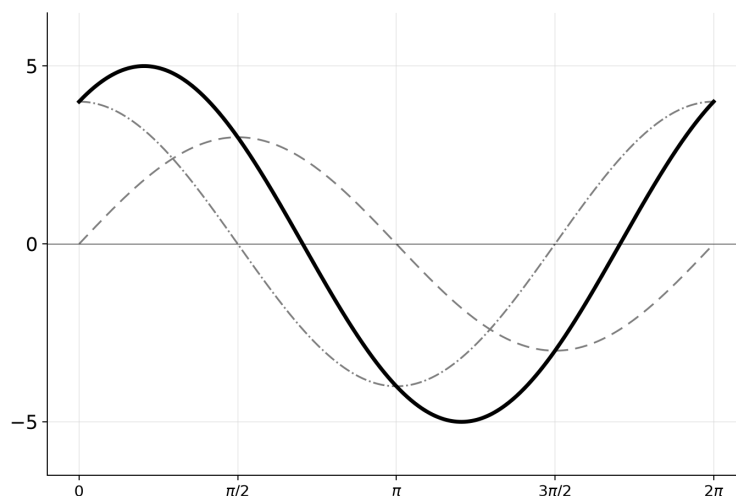
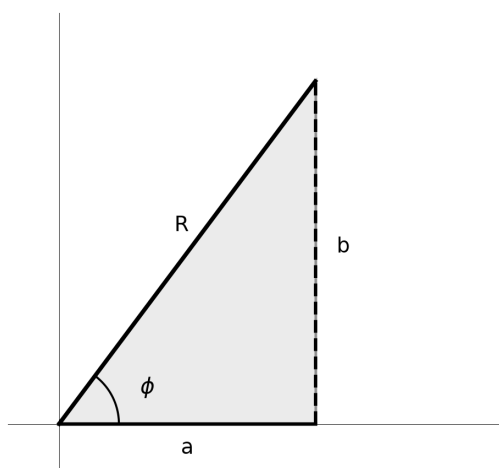
$$(1) R = \sqrt{a^2 + b^2} = \sqrt{9 + 16} = 5.$$

$$(2) \cos \phi = \frac{a}{R} = \frac{3}{5}, \sin \phi = \frac{b}{R} = \frac{4}{5}. \phi = \arcsin \frac{4}{5} \approx 53.13^\circ.$$

$$(3) 3 \sin x + 4 \cos x = 5 \sin(x + \phi).$$

$$\sqrt{3} \sin x - \cos x = 2 \sin(x - \frac{\pi}{6}). \sin x + \cos x = \sqrt{2} \sin(x + \frac{\pi}{4}).$$

$$\text{General form: } a \sin x \pm b \cos x = R \sin(x \pm \phi), a \cos x \pm b \sin x = R \cos(x \mp \phi).$$



Geometric insight: The coefficients (a, b) form a right triangle. The hypotenuse $R = \sqrt{a^2 + b^2}$ is the amplitude, angle ϕ is the phase shift. Left: phasor triangle. Right: two component waves merge into one shifted sine wave.

Example 4: Product-to-Sum and Sum-to-Product

$$\sin A \cos B = \frac{1}{2}[\sin(A + B) + \sin(A - B)].$$

$$\cos A \cos B = \frac{1}{2}[\cos(A + B) + \cos(A - B)].$$

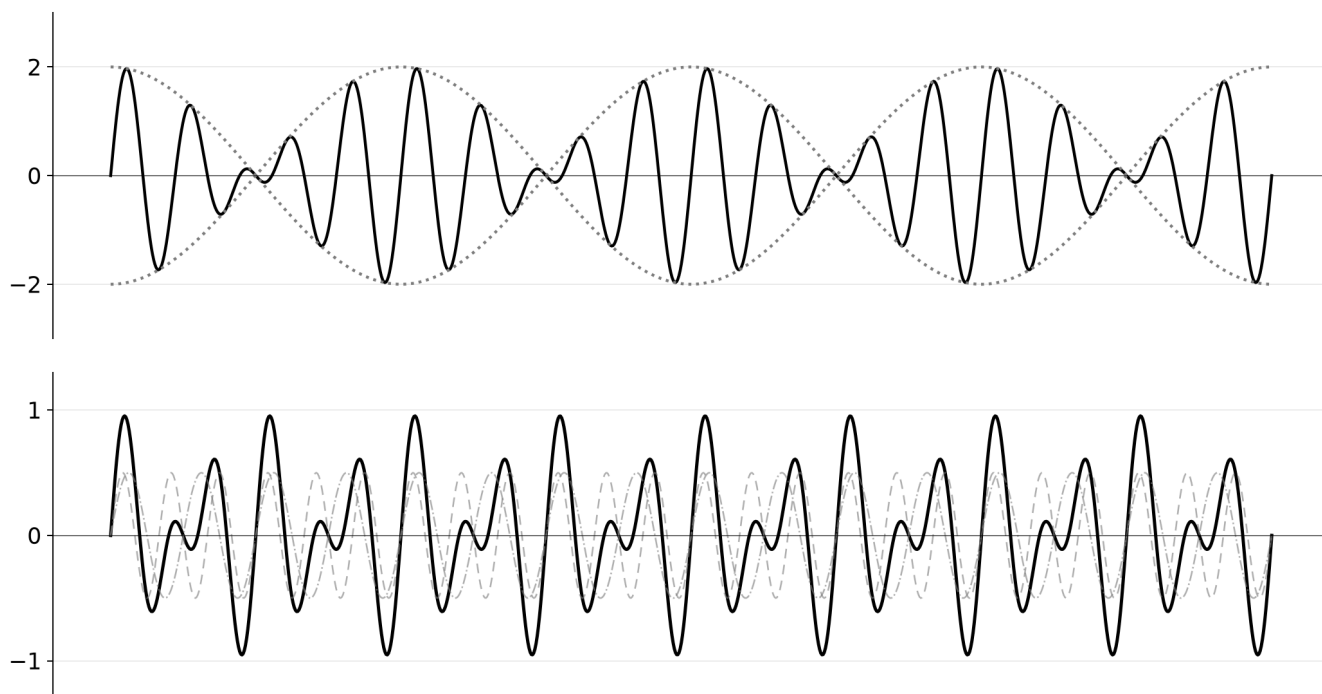
$$\sin A \sin B = \frac{1}{2}[\cos(A - B) - \cos(A + B)].$$

$$\sin A + \sin B = 2 \sin \frac{A+B}{2} \cos \frac{A-B}{2}.$$

$$\sin A - \sin B = 2 \cos \frac{A+B}{2} \sin \frac{A-B}{2}.$$

$$\cos A + \cos B = 2 \cos \frac{A+B}{2} \cos \frac{A-B}{2}.$$

$$\cos A - \cos B = -2 \sin \frac{A+B}{2} \sin \frac{A-B}{2}.$$



Geometric insight: Top — two close frequencies sum to a "beat" pattern (envelope: $2 \cos(\frac{A-B}{2}x)$). Bottom — $\sin A \cos B$ decomposes into two sine waves at $A + B$ and $A - B$.

Up to here: Sum/difference, double/half/triple angle, harmonic addition, product $\leftrightarrow$ sum. These identities are your tools for simplifying, solving, and proving.

Part B: Trigonometric Equations and Triangles

Example 5: Trigonometric Equations — Base Solution + $n \times$ Period

$$\sin x = \frac{1}{2} \rightarrow x = \frac{\pi}{6} + 2n\pi \text{ or } \frac{5\pi}{6} + 2n\pi.$$

$$2 \cos^2 x - \cos x - 1 = 0.$$

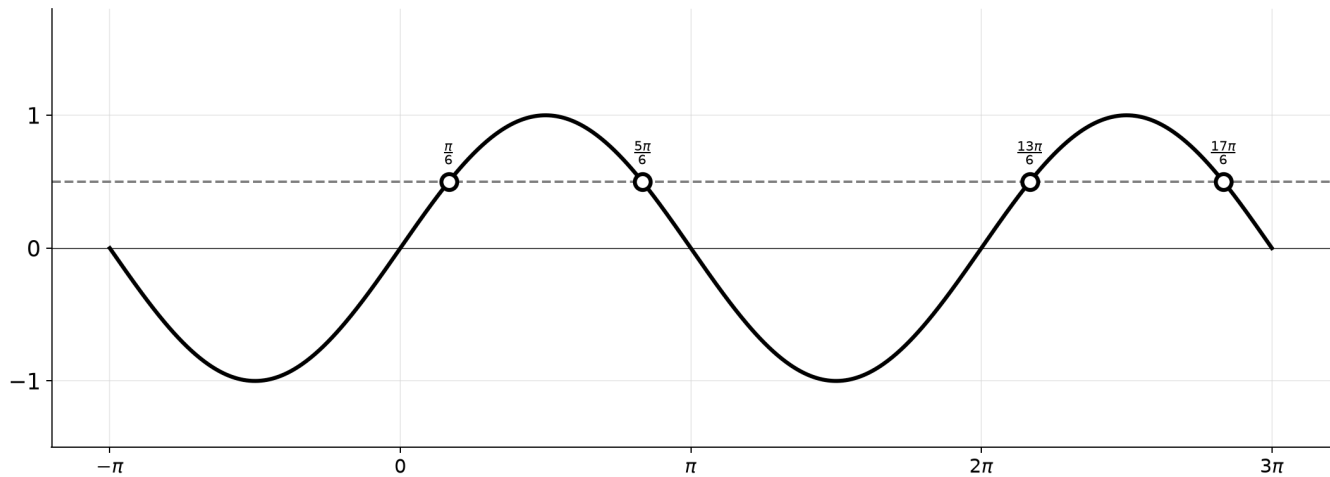
$$(1) t = \cos x: 2t^2 - t - 1 = 0 \rightarrow (2t + 1)(t - 1) = 0.$$

$$(2) \cos x = 1 \rightarrow x = 2n\pi. \cos x = -\frac{1}{2} \rightarrow x = \frac{2\pi}{3} + 2n\pi \text{ or } \frac{4\pi}{3} + 2n\pi.$$

$$\sin 2x = \cos x.$$

$$(1) 2 \sin x \cos x - \cos x = 0 \rightarrow \cos x(2 \sin x - 1) = 0.$$

$$(2) \cos x = 0 \rightarrow x = \frac{\pi}{2} + n\pi. \sin x = \frac{1}{2} \rightarrow x = \frac{\pi}{6} + 2n\pi \text{ or } \frac{5\pi}{6} + 2n\pi.$$



Geometric insight: $\sin x = \frac{1}{2}$ means finding where the sine wave meets $y = \frac{1}{2}$.
Intersections repeat every 2π , two per period.

Example 6: Law of Sines

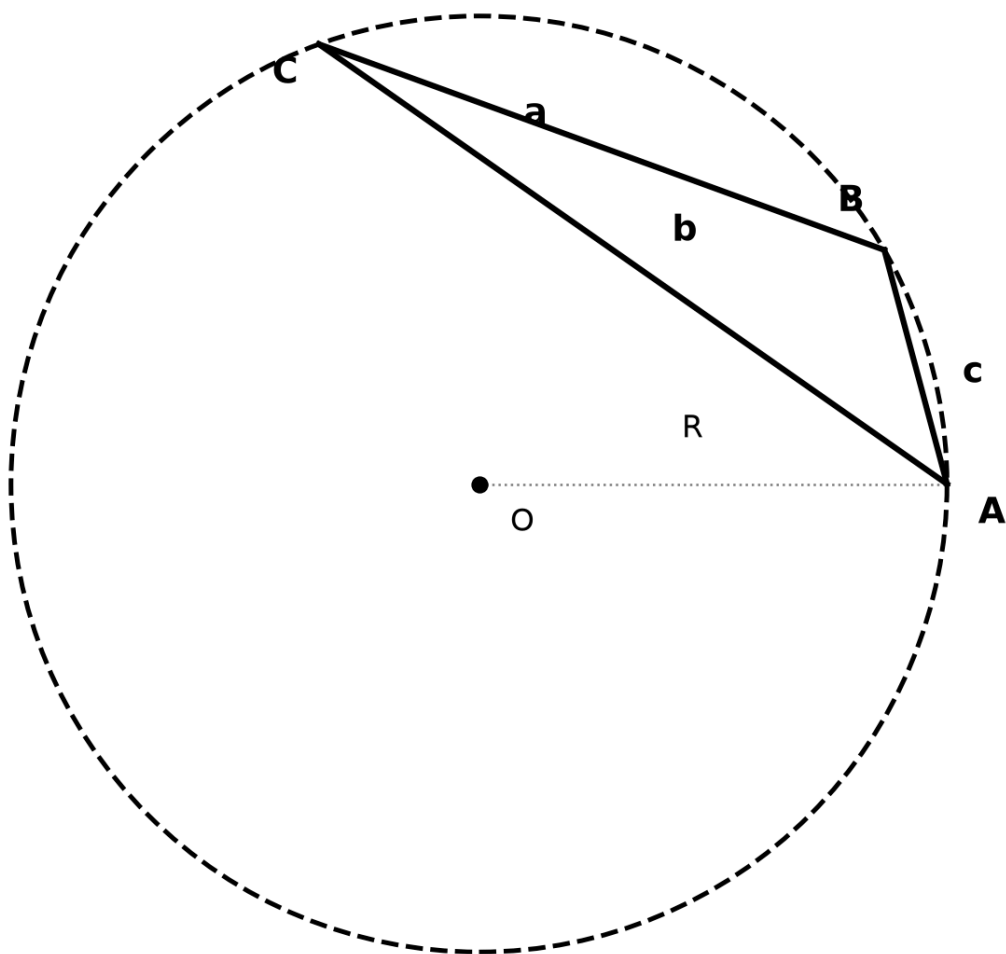
$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R.$$

Use when given two angles and one side (AAS/ASA).

$$a = 10, A = 30^\circ, B = 45^\circ \rightarrow b = a \frac{\sin B}{\sin A} = 10 \frac{\sin 45^\circ}{\sin 30^\circ} = 10\sqrt{2}.$$

Warning: Two sides and a non-included angle (SSA) may have 2 solutions (ambiguous case).

$$a = 5, b = 8, A = 30^\circ \rightarrow \sin B = \frac{8 \sin 30^\circ}{5} = 0.8 \rightarrow B \approx 53.1^\circ \text{ or } 126.9^\circ.$$



Geometric insight: Every triangle fits in a unique circumcircle. Each side is proportional to the sine of its opposite angle — constant of proportionality = $2R$ (the diameter). Longest side faces largest angle.

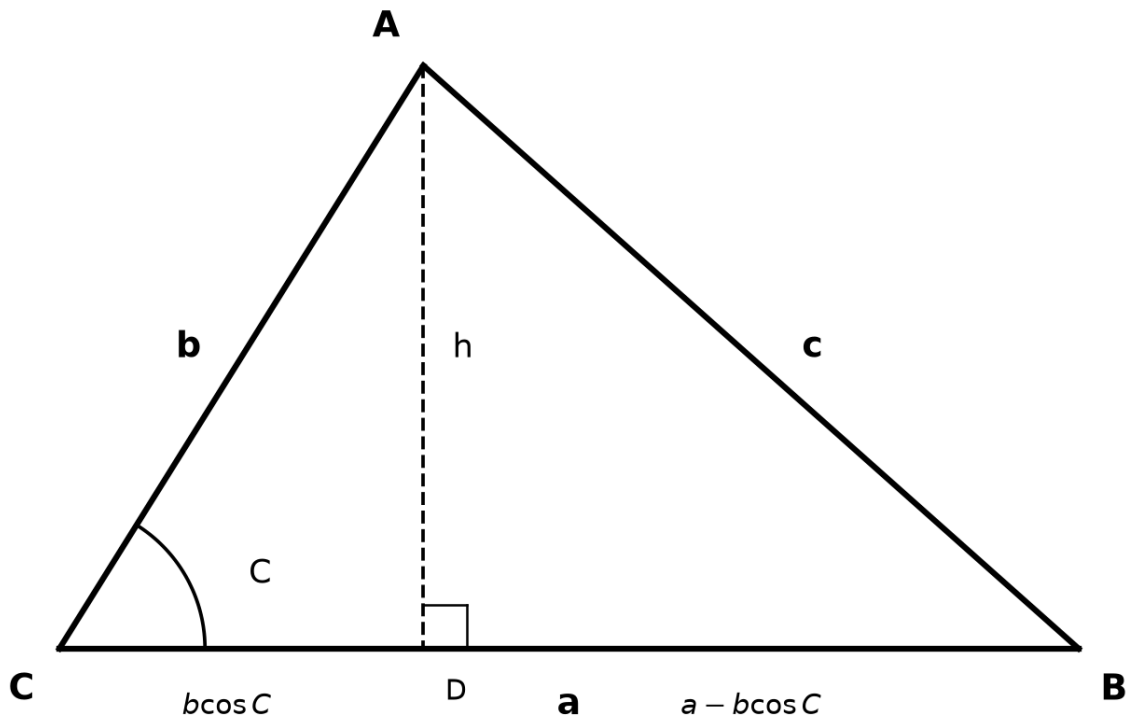
Example 7: Law of Cosines

$$a^2 = b^2 + c^2 - 2bc \cos A.$$

Two sides + included angle (SAS) $\rightarrow$ third side. Three sides (SSS) $\rightarrow$ any angle.

$$b = 5, c = 7, A = 60^\circ \rightarrow a^2 = 25 + 49 - 2 \cdot 5 \cdot 7 \cdot \frac{1}{2} = 39 \rightarrow a = \sqrt{39}.$$

$$\text{Triangle } 3,4,5 \rightarrow \cos A = \frac{4^2+5^2-3^2}{2 \cdot 4 \cdot 5} = \frac{4}{5}. A \approx 36.87^\circ.$$



Geometric insight: Drop altitude h from A to side a . Base splits into $b \cos C$ and $a - b \cos C$. Pythagorean theorem on the right triangle with hypotenuse c : $c^2 = h^2 + (a - b \cos C)^2 = b^2 \sin^2 C + a^2 - 2ab \cos C + b^2 \cos^2 C = a^2 + b^2 - 2ab \cos C$.

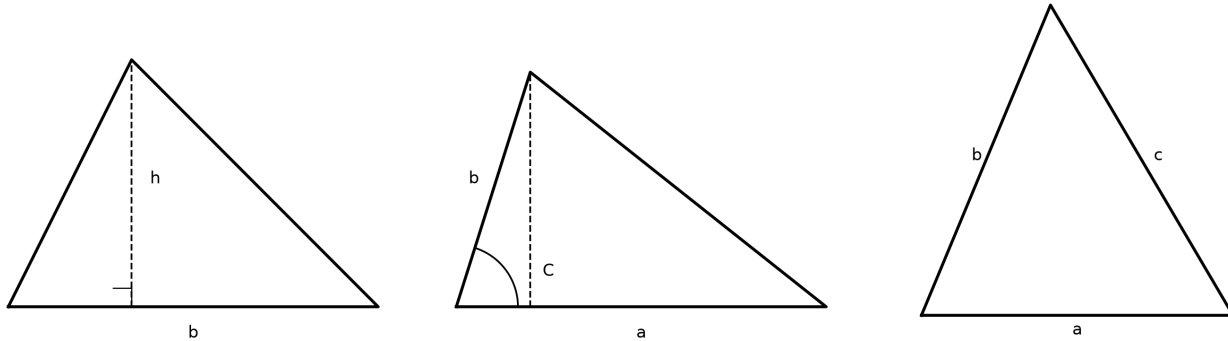
Example 8: Triangle Area — Three Ways

Two sides + included angle: Area = $\frac{1}{2}ab \sin C$.

$$a = 5, b = 8, C = 30^\circ \rightarrow \frac{1}{2} \cdot 5 \cdot 8 \cdot \sin 30^\circ = 10.$$

Heron's formula (three sides): $s = \frac{a+b+c}{2}$, Area = $\sqrt{s(s-a)(s-b)(s-c)}$.

$$3, 4, 5 \rightarrow s = 6. \sqrt{6 \cdot 3 \cdot 2 \cdot 1} = 6.$$



Geometric insight: (1) base $\times$ height. (2) height = $b \sin C$, same triangle, different inputs.
(3) Heron needs only side lengths — no angles.

Example 9: Inverse Trigonometric Functions

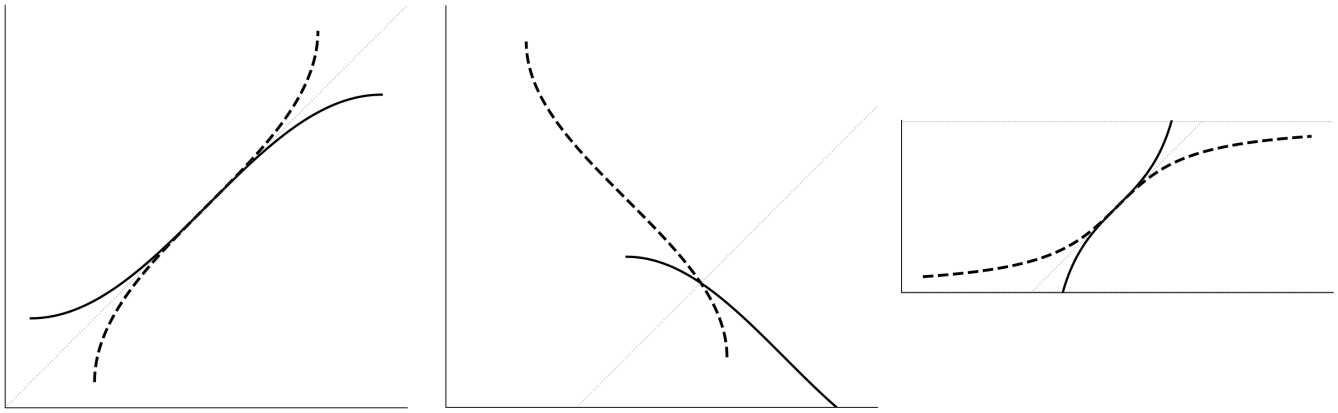
$$\arcsin x: \theta \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]. \arcsin \frac{1}{2} = \frac{\pi}{6}.$$

$$\arccos x: \theta \in [0, \pi]. \arccos \frac{1}{2} = \frac{\pi}{3}.$$

$$\arctan x: \theta \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right). \arctan 1 = \frac{\pi}{4}.$$

$$\arcsin(\sin \frac{5\pi}{6}): \sin \frac{5\pi}{6} = \frac{1}{2}. \arcsin \frac{1}{2} = \frac{\pi}{6}.$$

$\frac{5\pi}{6}$ lies outside arcsin's range; the answer is the value in $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ with the same sine.



Geometric insight: Inverse = reflect across $y = x$. To keep it a function, restrict domain:
 $\sin \rightarrow [-\frac{\pi}{2}, \frac{\pi}{2}]$, $\cos \rightarrow [0, \pi]$, $\tan \rightarrow (-\frac{\pi}{2}, \frac{\pi}{2})$.

Part C: Advanced Techniques — Beyond the Textbook

Example 10: Euler's Formula — The Deepest Bridge in Mathematics

$$e^{i\theta} = \cos \theta + i \sin \theta.$$

From this single equation, **every trig identity** follows algebraically:

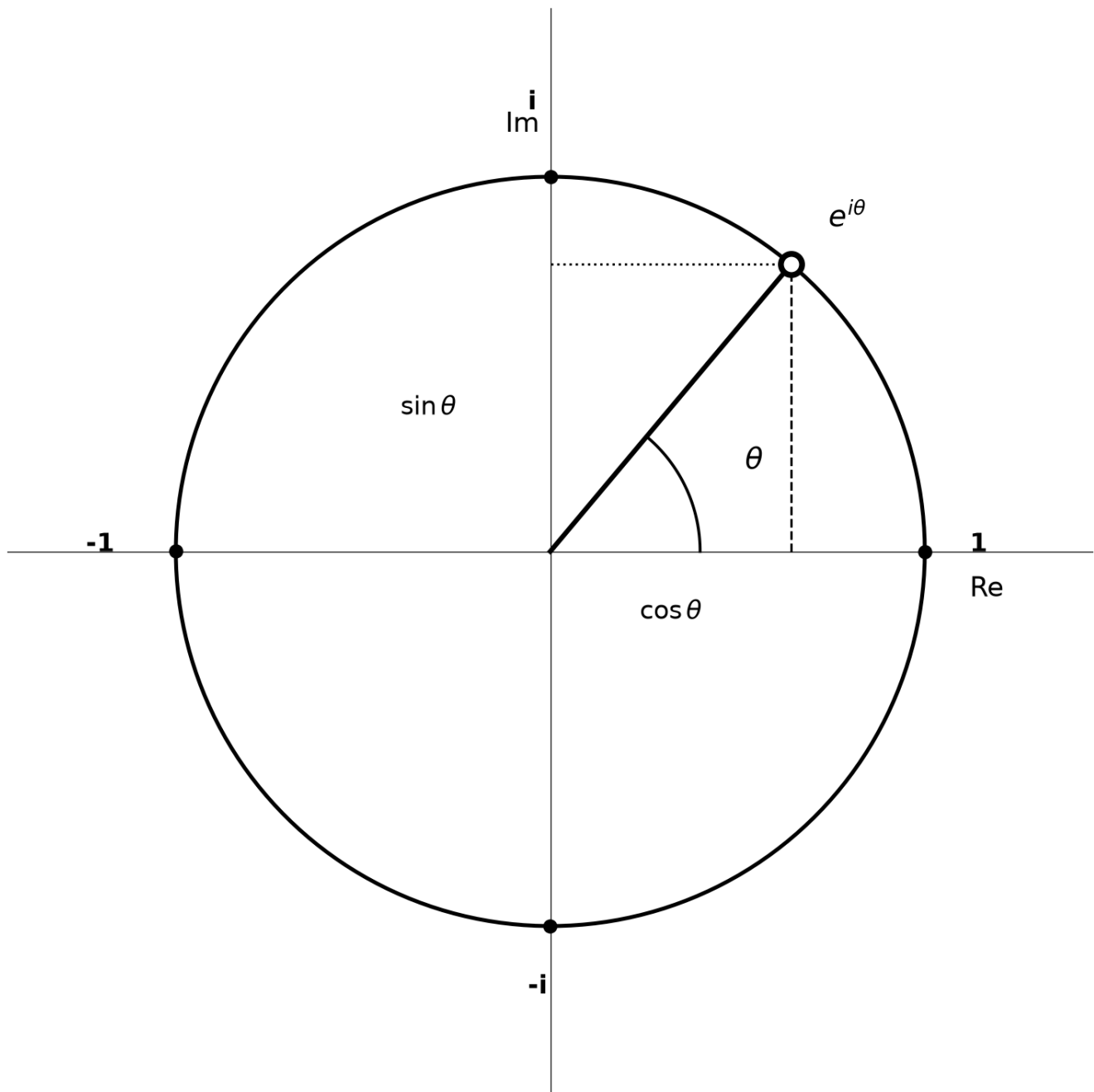
Deriving $\cos 2\theta$ via cubing:

$$e^{i2\theta} = (e^{i\theta})^2 = (\cos \theta + i \sin \theta)^2 = (\cos^2 \theta - \sin^2 \theta) + i(2 \sin \theta \cos \theta).$$

Equating real and imaginary parts: $\cos 2\theta = \cos^2 \theta - \sin^2 \theta$, $\sin 2\theta = 2 \sin \theta \cos \theta$.

De Moivre: $(\cos \theta + i \sin \theta)^n = \cos(n\theta) + i \sin(n\theta)$.

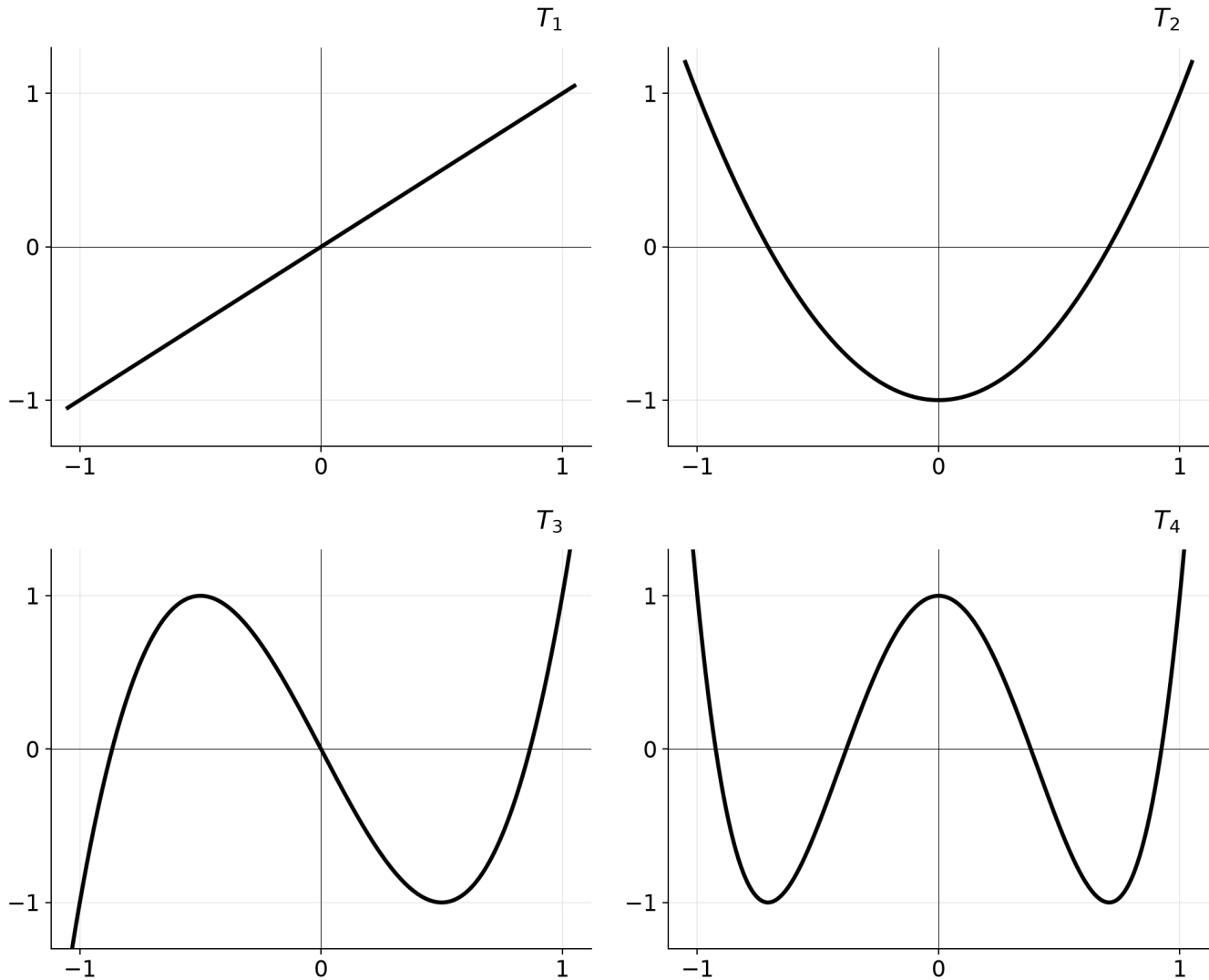
Roots of unity: $z^n = 1 \rightarrow z_k = e^{2\pi i k/n} = \cos \frac{2\pi k}{n} + i \sin \frac{2\pi k}{n}$, $k = 0, 1, \dots, n-1$.



Geometric insight: $e^{i\theta}$ rotates a point on the unit circle by θ . Real part = $\cos \theta$, imaginary = $\sin \theta$. De Moivre: $(e^{i\theta})^n = e^{in\theta}$ = rotate n times.

Example 11: Chebyshev Polynomials — Trig as Polynomials

$T_n(\cos \theta) = \cos(n\theta)$. $T_0(x) = 1$, $T_1(x) = x$, $T_2(x) = 2x^2 - 1$, $T_3(x) = 4x^3 - 3x$, $T_4(x) = 8x^4 - 8x^2 + 1$.



Geometric insight: On $[-1, 1]$, Chebyshev polynomials oscillate between -1 and 1 with $n + 1$ equally spaced extrema. Defined by $T_n(\cos \theta) = \cos(n\theta)$ — cosines disguised as polynomials.

To solve $\cos 3\theta = \frac{1}{2}$, let $x = \cos \theta$. Then $4x^3 - 3x = \frac{1}{2} \rightarrow 8x^3 - 6x - 1 = 0$.

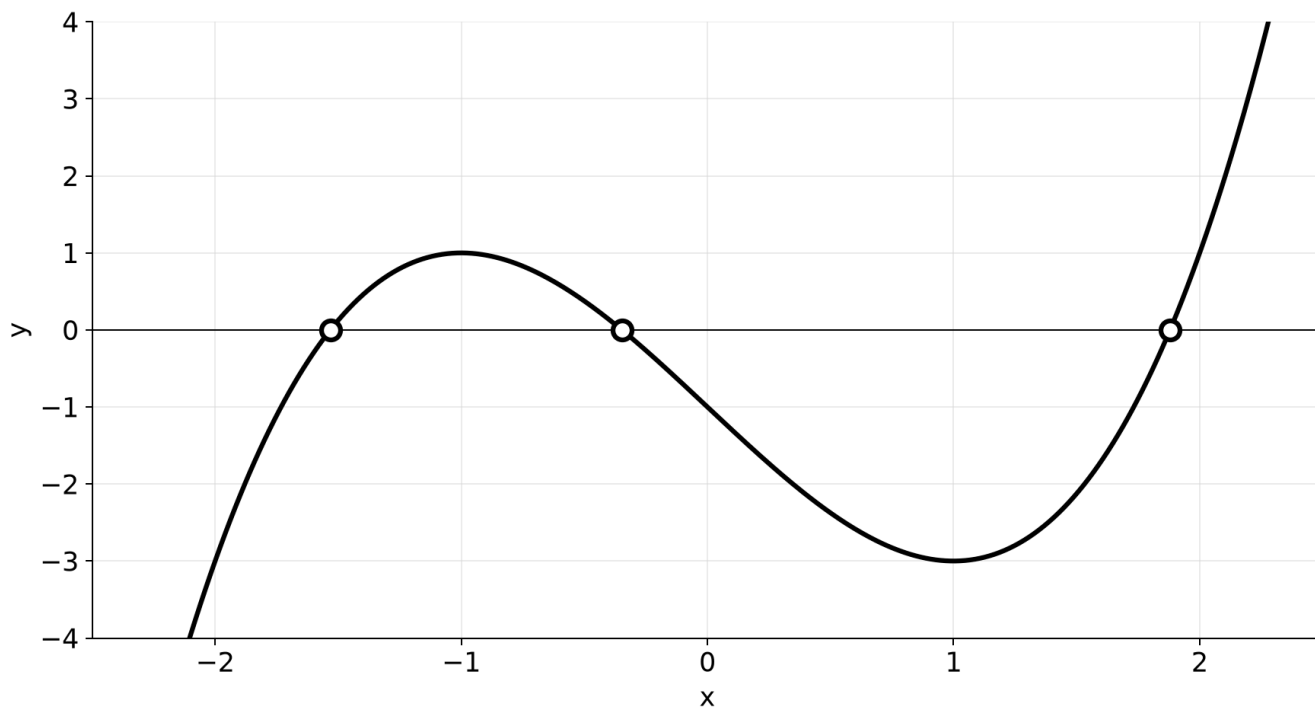
The roots are $\cos \frac{\pi}{9}$, $\cos \frac{7\pi}{9}$, $\cos \frac{13\pi}{9}$.

Example 12: The Cubic Equation via Trigonometry (Casus Irreducibilis)

For $x^3 + px + q = 0$ with three real roots, set $x = 2\sqrt{-\frac{p}{3}} \cos \theta$ where $\cos 3\theta = \frac{3q}{2p} \sqrt{-\frac{3}{p}}$.

Example: $x^3 - 3x - 1 = 0$. $p = -3$, $q = -1$. $2\sqrt{1} = 2$.

$\cos 3\theta = \frac{1}{2}$. Roots: $2 \cos \frac{\pi}{9}$, $2 \cos \frac{7\pi}{9}$, $2 \cos \frac{13\pi}{9}$.



Geometric insight: When a cubic has three real roots (casus irreducibilis), real radicals don't suffice — you need trig. Substituting $x = 2\sqrt{-p/3} \cos \theta$ turns $x^3 + px + q = 0$ into $\cos 3\theta = c$. Three roots = three angles solving $\cos 3\theta = c$ in $[0, \pi]$.

Example 13: Morrie's Law and Angle Product Magic

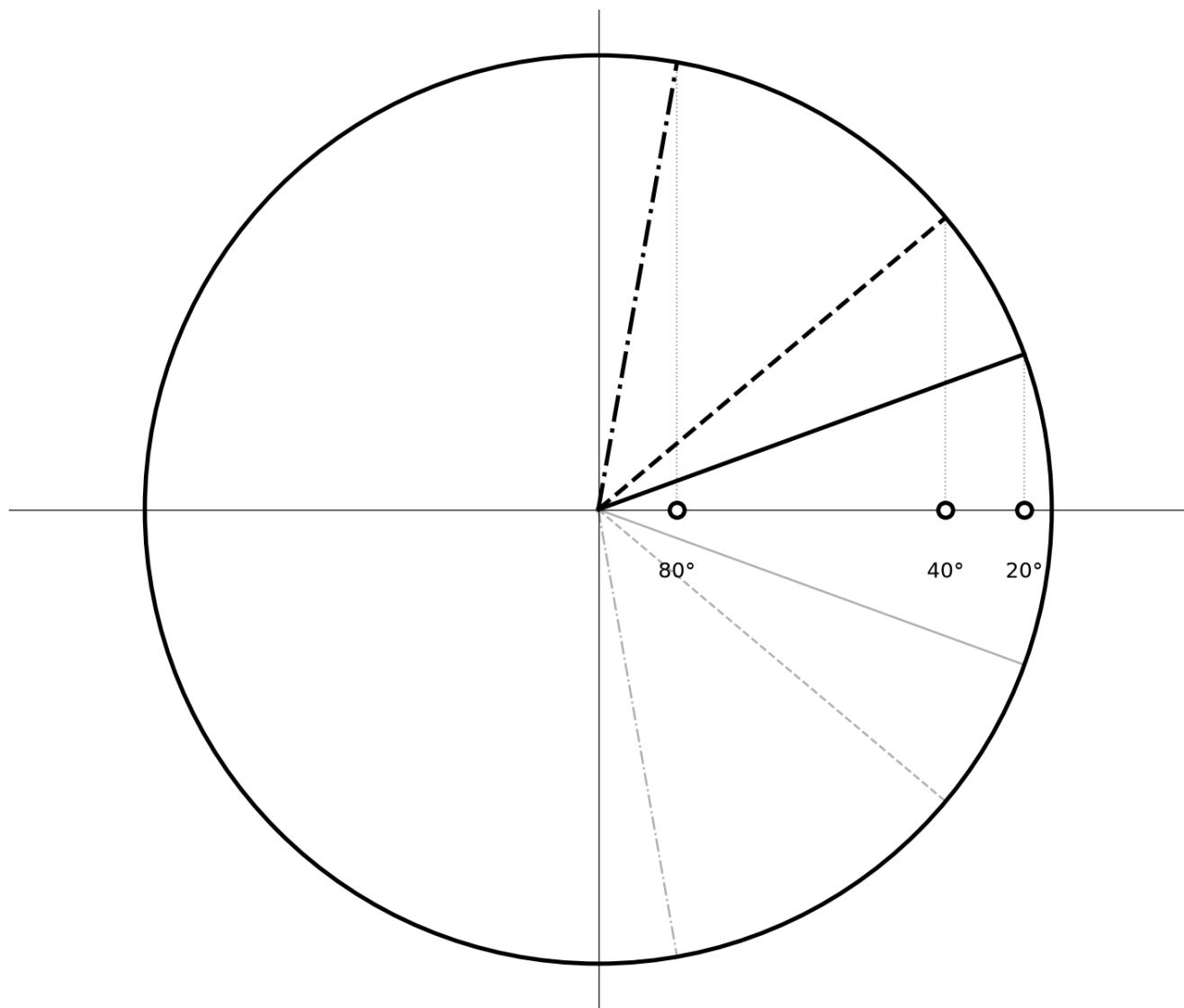
Morrie's Law: $\cos 20^\circ \cdot \cos 40^\circ \cdot \cos 80^\circ = \frac{1}{8}$.

Proof: multiply and divide by $\sin 20^\circ$, apply $\sin 2\theta = 2 \sin \theta \cos \theta$ repeatedly. Everything

cancels except $\frac{1}{8}$.

Generalization: $\prod_{k=1}^n \cos\left(\frac{\theta}{2^k}\right) = \frac{\sin \theta}{2^n \sin(\theta/2^n)}$.

Also: $\cos \frac{\pi}{7} \cos \frac{2\pi}{7} \cos \frac{3\pi}{7} = \frac{1}{8} \cdot \sin 10^\circ \sin 50^\circ \sin 70^\circ = \frac{1}{8}$.



Geometric insight: Each cosine = x -coordinate on unit circle at that angle. Proof: multiply by $\sin 20^\circ / \sin 20^\circ$, apply $\sin 2\theta = 2 \sin \theta \cos \theta$ — product telescopes to $\frac{1}{8}$.

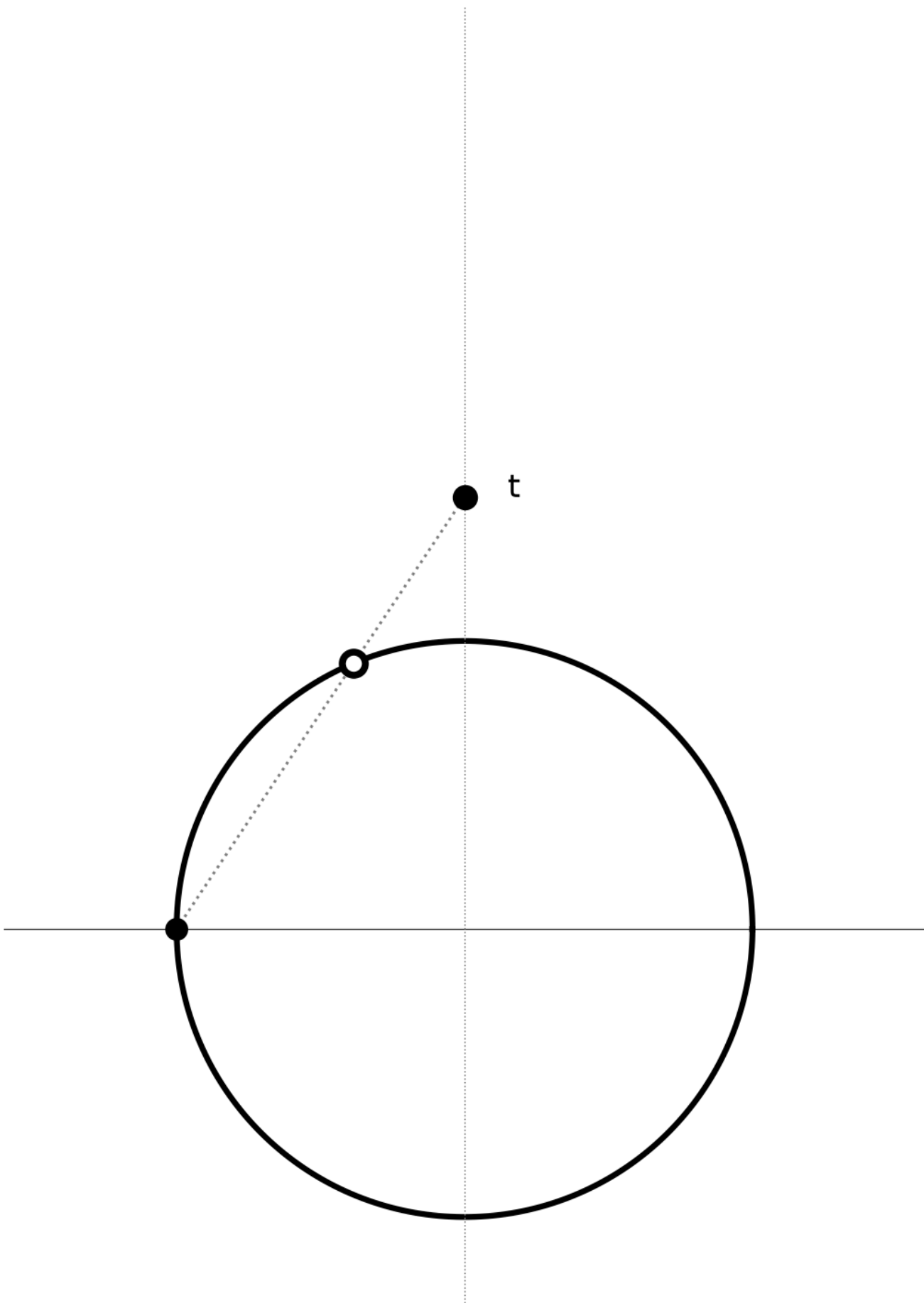
Example 14: Tangent Half-Angle Substitution (Weierstrass)

$t = \tan \frac{\theta}{2}$ turns any rational trig expression into a rational function:

$$\sin \theta = \frac{2t}{1+t^2}, \cos \theta = \frac{1-t^2}{1+t^2}, \tan \theta = \frac{2t}{1-t^2}.$$

Solving $\sin \theta + \cos \theta = 1$: Substitute $\rightarrow 2t + 1 - t^2 = 1 + t^2 \rightarrow t = 0$ or $t = 1$.

$\rightarrow \theta = 0$ or $\theta = \frac{\pi}{2}$ (plus multiples of 2π).

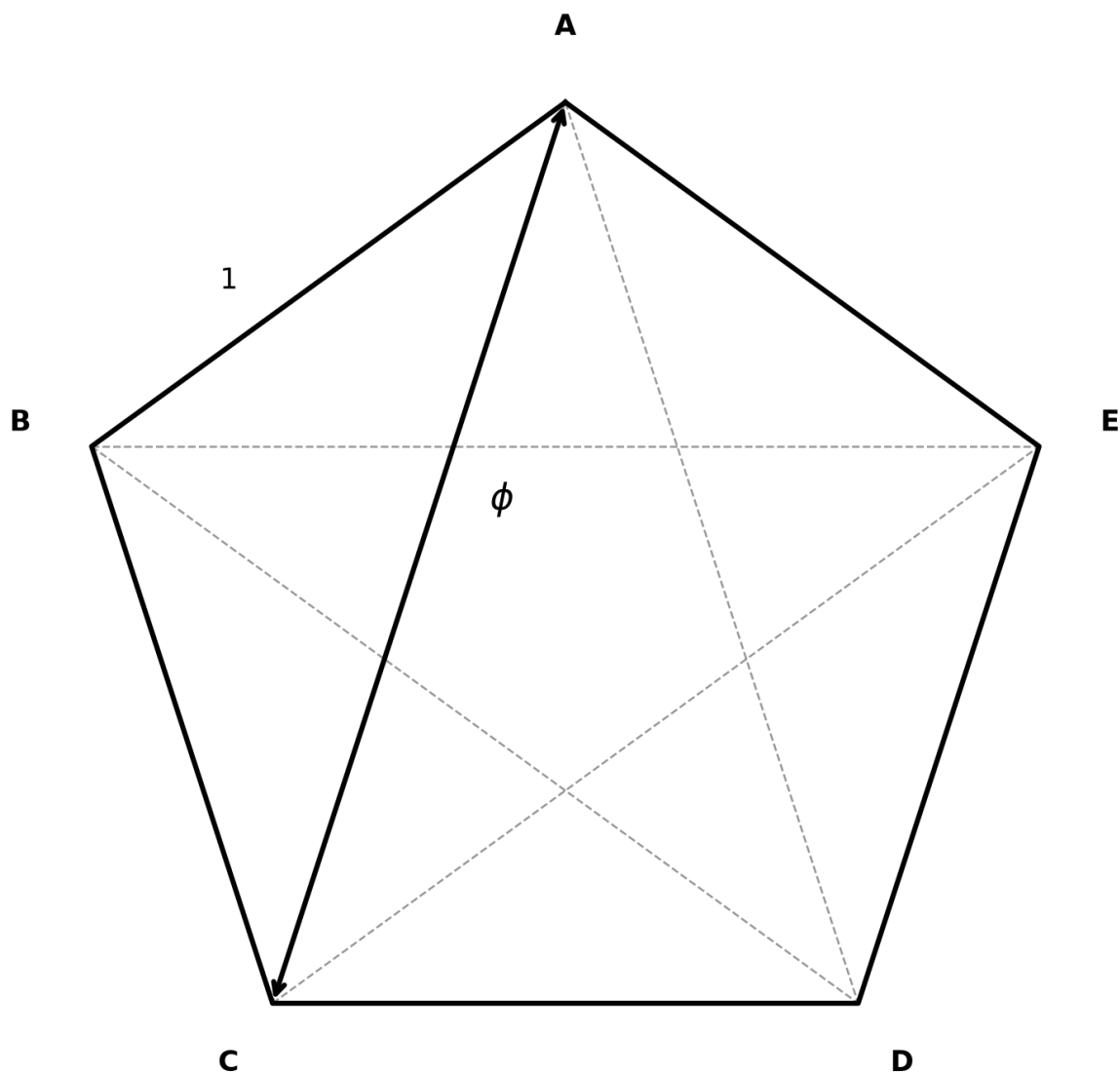


Geometric insight: $t = \tan(\theta/2)$ is where the line from $(-1, 0)$ through $(\cos \theta, \sin \theta)$ hits the y -axis — stereographic projection. Rational $t \leftrightarrow$ rational point on circle. This is why t -substitution turns trig integrals into rational ones.

Example 15: Sine of 18° and 36° — The Golden Ratio

$$\sin 18^\circ = \frac{\sqrt{5}-1}{4} \approx 0.3090. \cos 36^\circ = \frac{\sqrt{5}+1}{4} = \frac{\phi}{2} \approx 0.8090.$$

From the regular pentagon, the diagonal-to-side ratio is $\phi = \frac{1+\sqrt{5}}{2}$. Every integer degree from 3° to 90° has an exact radical expression via chaining 15° , 18° , and 36° values.



Geometric insight: In a regular pentagon, diagonal/side = $\phi = \frac{1+\sqrt{5}}{2}$. From this geometry: $\sin 18^\circ = \frac{\sqrt{5}-1}{4}$, $\cos 36^\circ = \frac{\phi}{2}$. Chain with 15° values to get exact radicals for any multiple of 3° .

Example 16: Polar Curves — Roses, Cardioids, and More

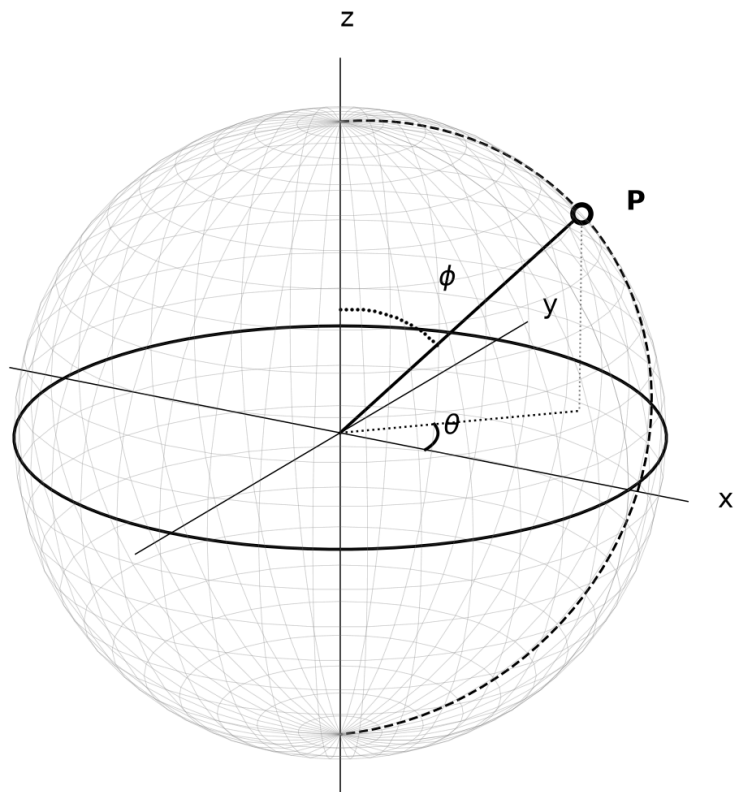
- **Rose curves:** $r = \cos(k\theta)$. k odd $\rightarrow k$ petals; k even $\rightarrow 2k$ petals.
- **Cardioid:** $r = 1 + \cos \theta$ (heart-shaped).
- **Limacon:** $r = a + b \cos \theta$ (inner loop when $|a| < |b|$).
- **Lemniscate:** $r^2 = \cos 2\theta$ (figure-eight).

Parametric conversion: $x = r(\theta) \cos \theta$, $y = r(\theta) \sin \theta$.

Visual Interlude: Trigonometry in 3D and Beyond

Spherical coordinates — polar coordinates extended to 3D. A point is (r, θ, ϕ) :

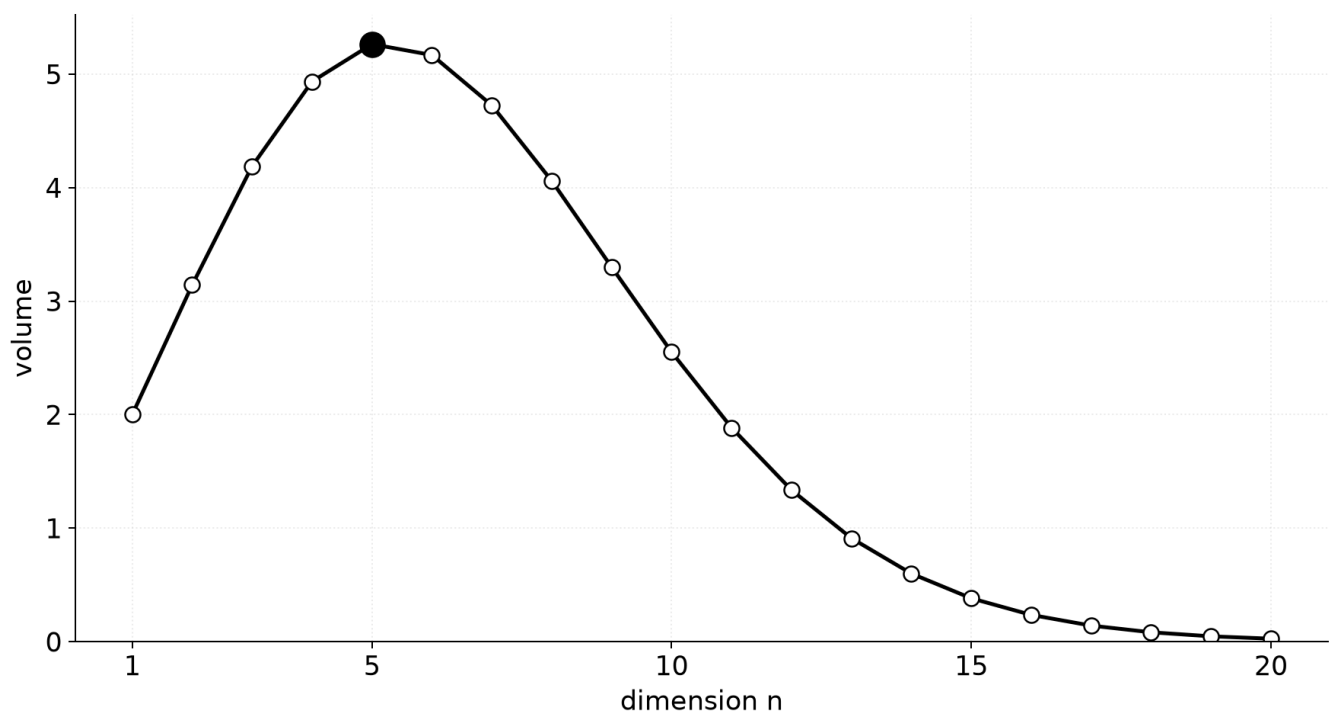
$$x = r \sin \phi \cos \theta, y = r \sin \phi \sin \theta, z = r \cos \phi.$$



The unit sphere ($r = 1$) is parametrized by sine and cosine alone. ϕ is the polar angle from the z -axis; θ is the azimuthal angle in the xy -plane.

Hyperspheres in n dimensions — the volume of an n -dimensional ball of radius R :

$$V_n(R) = \frac{\pi^{n/2}}{\Gamma(n/2+1)} R^n.$$



The surface area involves $\int \sin^{n-2} \phi \, d\phi$ — trig generalizes to every dimension. Spherical harmonics $Y_\ell^m(\theta, \phi)$, the higher-dimensional cousins of $\sin(k\theta)$ and $\cos(k\theta)$, power quantum mechanics, geophysics, and computer graphics.

Decision Tree — Trig Equations

You encounter a trigonometric equation:

- └─ (1) Only one type of trig function?
 - └─ Substitute $t = \sin x$ (or $\cos x$). $t \in [-1, 1]$. Solve polynomial.
- └─ (2) Different angles? ($2x$ vs x)
 - └─ Use double/triple/half-angle to unify all angles.
- └─ (3) $\sin$ and $\cos$ mixed?
 - └─ $\sin = \cos \rightarrow$ divide by $\cos$ (check $\cos \neq 0$).
 - └─ $a \sin + b \cos = c \rightarrow$ harmonic addition: $R \sin(x+\phi) = c$.
 - └─ Product = 0 $\rightarrow$ set each factor = 0.
- └─ (4) Rational in $\sin/\cos$?
 - └─ Tangent half-angle: $t = \tan(x/2)$.
- └─ (5) Restricted domain?
 - └─ Find general solution, pick n so answer falls in range.
- └─ (6) Inequality?
 - └─ Use unit circle or graph. Find intervals, apply period.

Decision Tree — Trig Inequalities

You encounter a trigonometric inequality:

- └─ (1) $\sin x > k$ or $\cos x > k$ form?
 - └─ Use unit circle. Mark the angular interval above/below height k . Apply period.
- └─ (2) Quadratic form? ($2 \sin^2 x - \sin x - 1 < 0$)
 - └─ t -substitution $\rightarrow t$ -range $\rightarrow \sin x$ range $\rightarrow x$ intervals.
- └─ (3) Product > 0 or < 0 ?
 - └─ Sign chart. Find zeros of each factor. Partition intervals.

Common Mistakes

Mistake 1: $\cos(A + B) = \cos A + \cos B$

Wrong path: " $\cos(A + B) = \cos A + \cos B$."

Why wrong: Cosine does not distribute over addition.

Right path: $\cos(A + B) = \cos A \cos B - \sin A \sin B$. You must multiply, not add.

Mistake 2: Dividing by $\cos x$ without checking $\cos x = 0$

Wrong path: " $\sin x = \cos x \rightarrow$ divide by $\cos x \rightarrow \tan x = 1$."

Why wrong: If $\cos x = 0$, the division is illegal. Solutions where $\cos x = 0$ might be lost.

Right path: Check $\cos x = 0$ first. For $\sin x = \cos x$, $\cos x = 0$ gives no solution. Then divide safely.

Mistake 3: Mismatching side and opposite angle in Law of Sines

Wrong path: Writing $\frac{a}{\sin B}$.

Why wrong: Side a sits opposite angle A . Each side pairs with its own opposite angle.

Right path: $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$. Match the letters.

What We Just Did

- (1) Identity toolkit – sum/difference (6 formulas). Double/half/triple angle. Harmonic addition merges a $\sin + b \cos$ into one wave. Product $\leftrightarrow$ sum converts multiplication to addition and vice versa.
- (2) Equation solving – one trig type $\rightarrow$ t-sub. Mixed angles $\rightarrow$ unify via double-angle. Mixed $\sin + \cos \rightarrow$ factor or harmonic-add. Rational trig $\rightarrow$ tan half-angle. Triangles: SAS $\rightarrow$ law of cosines, AAS $\rightarrow$ law of sines. Area = $\frac{1}{2}ab \sin C$ or Heron.
- (3) Advanced – Euler's formula unifies all identities. Chebyshev polynomials encode $\cos(n\theta)$. Cubic equations solved via trig. Morrie's Law and product gems. Golden ratio hides in $\sin 18^\circ$. Spherical coordinates extend trig to 3D. Hypersphere volumes use trig in n dimensions.

Practice 1

Solve $\cos 2x = \sin x$ on $[0, 2\pi]$. Use $\cos 2x = 1 - 2\sin^2 x$.

$\rightarrow$ Reference: **Example 2, 5**

Solutions: [Solutions](#)

Practice 2: Composition

Write $5 \sin x + 12 \cos x$ as $R \sin(x + \phi)$. Find the maximum value and the x at which it occurs.

$\rightarrow$ Reference: **Example 3**

Solutions: [Solutions](#)

Practice 3

Triangle with $a = 7$, $b = 10$, $c = 13$. Find all three angles and the area.

→ Reference: **Example 7, 8**

Solutions: [Solutions](#)

Practice 4: Real Battle

$\sec x + \tan x = 2$. Find $\sec x - \tan x$ and $\sin x$.

Use $(\sec x + \tan x)(\sec x - \tan x) = 1$.

→ Reference: **Pythagorean identity**

Solutions: [Solutions](#)

Practice 5

Solve $\sin 3x = \cos x$ on $[0, 2\pi]$.

Hint: $\sin 3x = \cos x = \sin(\frac{\pi}{2} - x)$.

→ Reference: **Example 1, 5**

Solutions: [Solutions](#)

Practice 6: Composition

Using Euler's formula $e^{i\theta} = \cos \theta + i \sin \theta$, derive formulas for $\sin 3\theta$ and $\cos 3\theta$ by cubing $e^{i\theta}$.

→ Reference: **Example 10**

Solutions: [Solutions](#)

Practice 7

Solve $x^3 - 3x + 1 = 0$ using the trigonometric method. Identify the three real roots as $2 \cos \alpha$, $2 \cos \beta$, $2 \cos \gamma$.

→ Reference: **Example 12**

Solutions: [Solutions](#)

Practice 8: Real Battle

Find the exact value of $\sin 3^\circ$ as a radical expression using $\sin 3^\circ = \sin(18^\circ - 15^\circ)$.

→ Reference: **Example 15**

Solutions: [Solutions](#)

Basic Algebra Drill — Identities and Equations (10 Problems)

Pure calculation refresher. Rapid evaluation using identities.

D1. Evaluate $\sin 75^\circ$ using the sum formula $\sin(45^\circ + 30^\circ)$.

D2. Evaluate $\cos 105^\circ$ using the sum formula $\cos(60^\circ + 45^\circ)$.

D3. Evaluate $\tan 15^\circ$ using the difference formula.

D4. Given $\sin A = \frac{3}{5}$ (A in Q1) and $\cos B = \frac{5}{13}$ (B in Q1), find $\sin(A + B)$.

- D5.** Given $\cos \theta = -\frac{4}{5}$ (θ in Q2), find $\cos 2\theta$ and determine its sign.
- D6.** Write $\sin 3x \cdot \cos x$ as a sum of two sine terms (product-to-sum).
- D7.** Write $\sin 5x + \sin x$ as a product (sum-to-product).
- D8.** Evaluate $\arcsin(\frac{\sqrt{3}}{2})$ and $\arccos(-\frac{1}{2})$.
- D9.** Find the exact value of $\sin(\arcsin \frac{3}{5} + \arccos \frac{5}{13})$.
- D10.** In triangle ABC , $A = 40^\circ$, $B = 60^\circ$, $a = 8$. Find side b using the law of sines.

Solutions: [Solutions](#)

Advanced Algebra Drill — Trigonometric Identities and Equations (10 Problems)

Multi-step. Each problem chains 2–3 identities. Covers all major formula families.

- A1.** Simplify $\frac{\sin 2x}{1 + \cos 2x}$. Express as a single trig function.
- A2.** Express $\sin^4 \theta - \cos^4 \theta$ in terms of $\cos 2\theta$ only. (Hint: factor as difference of squares.)
- A3.** Prove $\frac{\sin 2x}{1 - \cos 2x} = \cot x$.
- A4.** In triangle ABC , $a = 8$, $b = 6$, $\angle C = 60^\circ$. Find side c and the area.
- A5.** Simplify $\frac{\cos 3\theta}{\cos \theta} + \frac{\sin 3\theta}{\sin \theta}$. (Use triple-angle formulas.)
- A6.** Compute $\sin 15^\circ \cdot \cos 15^\circ \cdot \cos 30^\circ$. Simplify to $\frac{\sqrt{m}}{n}$ form.
- A7.** Write $3 \sin x + \sqrt{3} \cos x$ in the form $R \sin(x + \phi)$. Give R and ϕ exactly.
- A8.** Solve $2 \sin^2 x - 3 \sin x + 1 = 0$ for $x \in [0, 2\pi]$.
- A9.** If $\tan \theta = \frac{3}{4}$ and $\theta \in Q3$, compute $\sin 2\theta$ and $\cos 2\theta$ exactly.
- A10.** In triangle ABC , $a = 7$, $b = 8$, $c = 9$. Find the largest angle (opposite the longest side) and the area.

Today's Procedure

- Step 1: Wield the identities – sum/difference splits angles. Double-angle bridges $2x \leftrightarrow x$.
Harmonic addition merges a $\sin + b \cos$ into $R \sin(x + \phi)$. Product $\leftrightarrow$ sum converts multiplication to addition. Use these to simplify, factor, and prove.
- Step 2: Solve equations – t-sub for single trig type. Unify mixed angles via double-angle
Factor when product=0. Harmonic-add for a $\sin + b \cos = c$. $\tan(x/2)$ as universal fal
Inequalities: draw the unit circle, mark the interval, repeat with period.
- Step 3: Extend – Euler's formula gives every identity algebraically. Cubic equations
solved via trig avoid complex numbers. Regular pentagon gives exact $\sin 18^\circ$.
Spherical coordinates extend trig to 3D. Hypersphere volumes use trig in nD .

How to Read These Symbols

Symbol	Reads as	Meaning
$\sin(A \pm B)$	"sine of A plus or minus B"	sine addition formula: $\sin A \cos B \pm \cos A \sin B$
$\cos(A \pm B)$	"cosine of A plus or minus B"	cosine addition formula: $\cos A \cos B \mp \sin A \sin B$
$\tan(A \pm B)$	"tangent of A plus or minus B"	$(\tan A \pm \tan B) / (1 \mp \tan A \tan B)$
$\sin^2 \theta = \frac{1 - \cos 2\theta}{2}$	"sine squared theta equals one minus cosine two theta over two"	power-reduction — used in integration
$\cos^2 \theta = \frac{1 + \cos 2\theta}{2}$	"cosine squared theta equals one plus cosine two theta over two"	power-reduction — used in integration

Symbol	Reads as	Meaning
$\sin^{-1} x$, $\cos^{-1} x$, $\tan^{-1} x$	"inverse sine of x" / "arcsine of x"	inverse trig — returns an angle
$\arcsin x$	"arcsine of x"	alternative notation for $\sin^{-1} x$ (avoids confusion with $1/\sin$)
Law of Sines	"Law of Sines"	$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$
Law of Cosines	"Law of Cosines"	$c^2 = a^2 + b^2 - 2ab \cos C$ — generalizes Pythagorean theorem
harmonic identity	"harmonic identity" / "auxiliary angle method"	$a \sin x + b \cos x = R \sin(x + \phi)$ where $R = \sqrt{a^2 + b^2}$

Terminology

Up to now we used plain words like "splitting angles", "harmonic addition", "golden ratio".

You have already learned all the methods. Now we attach the formal mathematical names.

What we called it	Mathematical term	Notation
sum/difference formulas	sum and difference identities	$\sin(A \pm B), \cos(A \pm B)$
double/triple/half-angle	multiple-angle formulas	$\sin 2\theta, \sin 3\theta$
harmonic addition	harmonic addition / auxiliary angle	$a \sin x + b \cos x = R \sin(x + \phi)$
product-to-sum	product-to-sum identities	$\sin A \cos B = \frac{1}{2}[\sin(A + B) + \sin(A - B)]$

What we called it	Mathematical term	Notation
law of sines	law of sines	$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$
law of cosines	law of cosines	$a^2 = b^2 + c^2 - 2bc \cos A$
Heron's formula	Heron's formula	$\sqrt{s(s-a)(s-b)(s-c)}$
inverse trig	inverse trigonometric functions	arcsin, arccos, arctan
Euler's formula	Euler's formula	$e^{i\theta} = \cos \theta + i \sin \theta$
Chebyshev polynomials	Chebyshev polynomials of the first kind	$T_n(\cos \theta) = \cos(n\theta)$
casus irreducibilis	casus irreducibilis	cubic with 3 real roots via trig
Weierstrass substitution	tangent half-angle substitution	$t = \tan(\theta/2)$
spherical coordinates	spherical coordinates	$(r, \theta, \phi), x = r \sin \phi \cos \theta$
hypersphere	n -sphere / hypersphere	$V_n = \pi^{n/2} R^n / \Gamma(n/2 + 1)$